

Modeling interception movement distributions

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A compact personal representation for stochastic movement generation

One project, two modeling tasks

Can a compact personal representation capture movement variability?

Reconstruction

Encode an observed trial and decode it back.

Does compression preserve its trajectory?

Generation

Use separate context trials to sample new movements.

Do their distributions match held-out recordings?

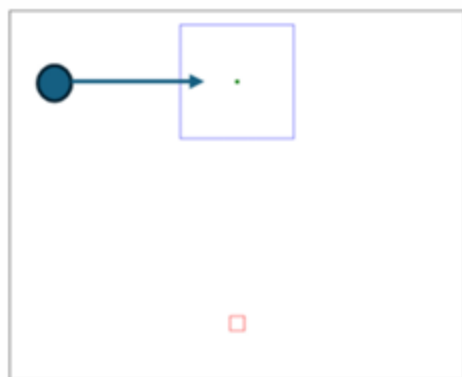
Main evaluation: generation. Reconstruction tests compression; personal-center controls test the value of personal context.

The task and the evaluation split

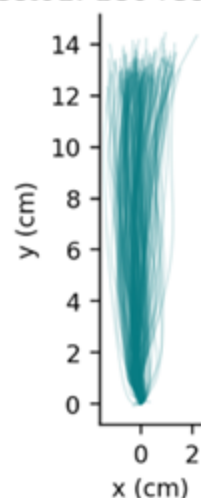
4,732 trials from 28 participants

Four rotations: 17 train / 4 validation / 7 test

Task display



subject01: 180 recorded trials



The eleven kinematic features

Timing

Response time; movement time; relative time to peak speed

Speed

Peak speed; speed-peak count

Geometry

Path length; straight-line distance; curvature; maximum lateral deviation

Endpoint

Final x and y

Eleven features summarize timing, speed and geometry. Coordinates are in cm. Every participant enters a test fold once.

The compression pipelines

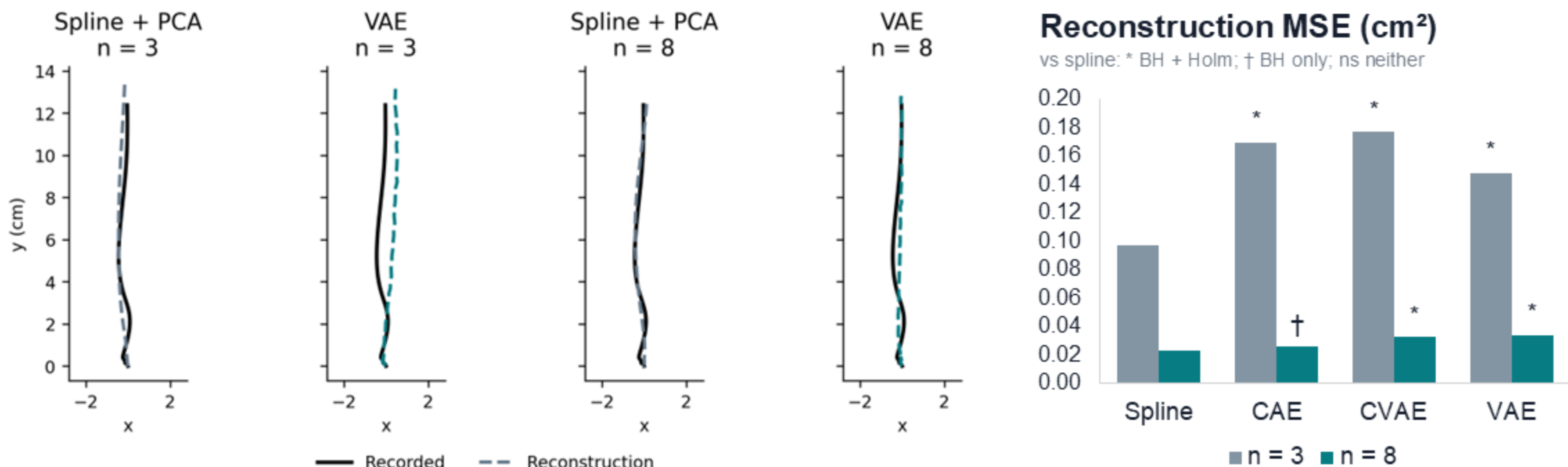
Every trajectory becomes 100 x 2 positions on normalized time.

Stage	Spline + PCA	Neural models
Input	100 x 2 positions	200 standardized coordinates
Encoder	9 coefficients per axis; PCA on N x 18 matrix	Two 256-unit layers; n-value code
Decoder	Inverse PCA; two spline basis products	Two 256-unit layers; 200 coordinates
Timing output	Ridge on code + 5 conditions	Separate two-duration head

VAE: variational autoencoder. CVAE adds task conditions; CAE uses a deterministic code. Matched n = 3 and 8.

Reconstruction favors spline + PCA

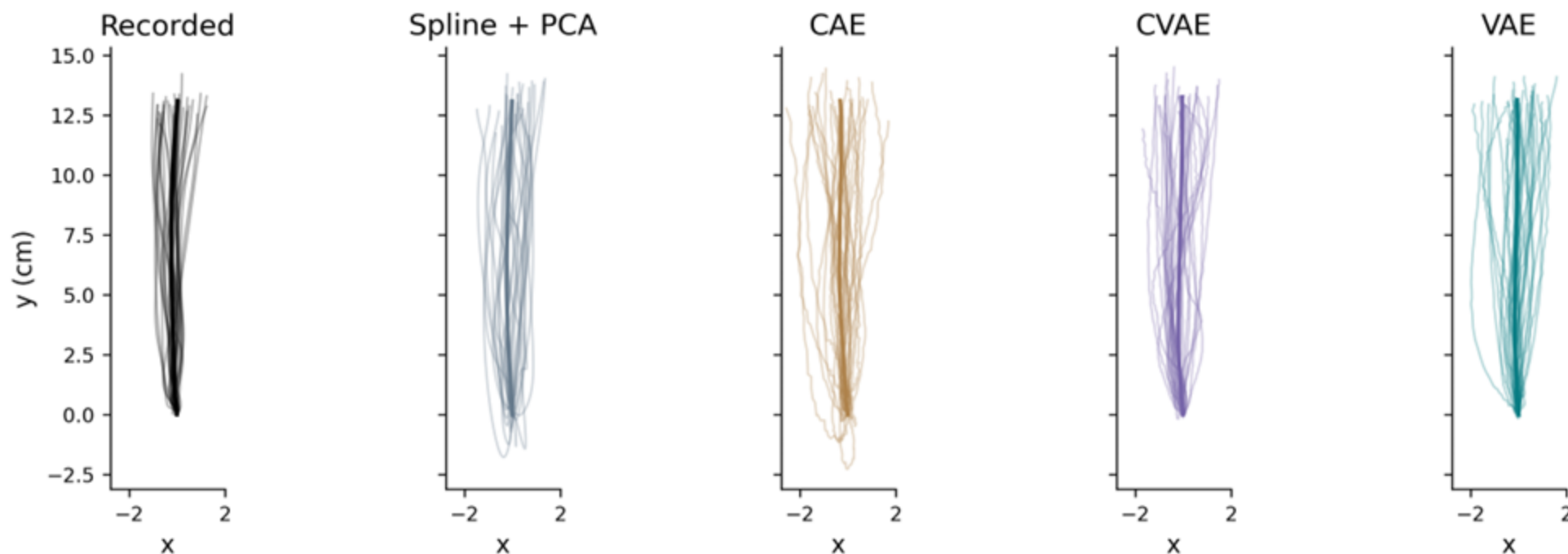
The encoder receives the observed test trajectory.



Black: recorded; dashed: reconstructed. One median-duration trial. Lower MSE is better. Spline beats each neural model under BH-140; spline vs CAE $n=8$ does not survive Holm.

Generation uses separate context trials

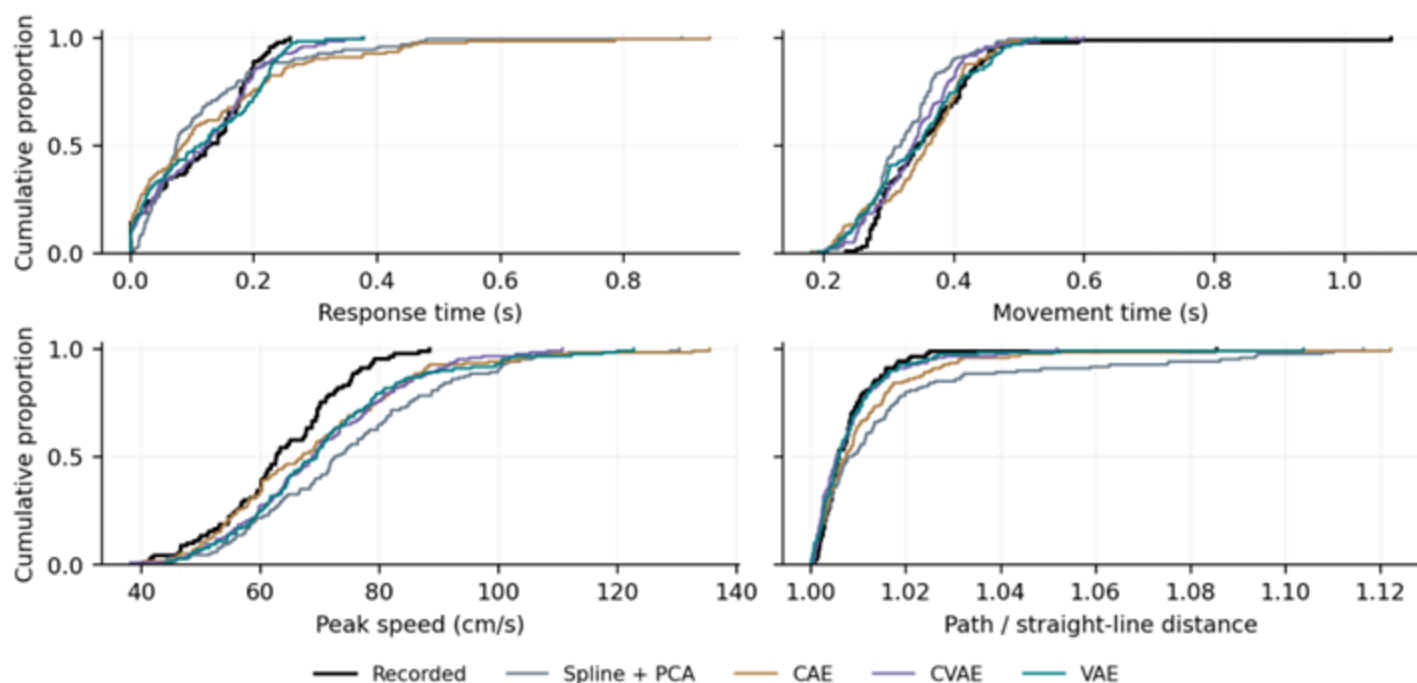
Context codes establish a personal center. We decode 120 draws around it using shared training covariance.



Same participant, separate query recordings. $n=8$, all models. Both capacities appear in the report. Thin: first 30 paths. Thick: mean of all paths. Equal x/y scale in cm.

Generated kinematic-feature distributions

Cumulative curves show individual features. Mean KS summarizes all eleven feature discrepancies.



Mean feature KS

vs spline: * BH + Holm; † BH only; ns neither

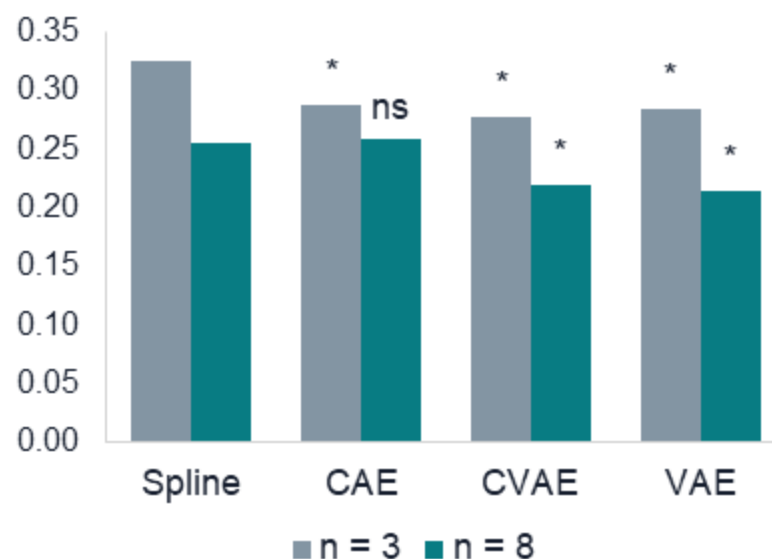


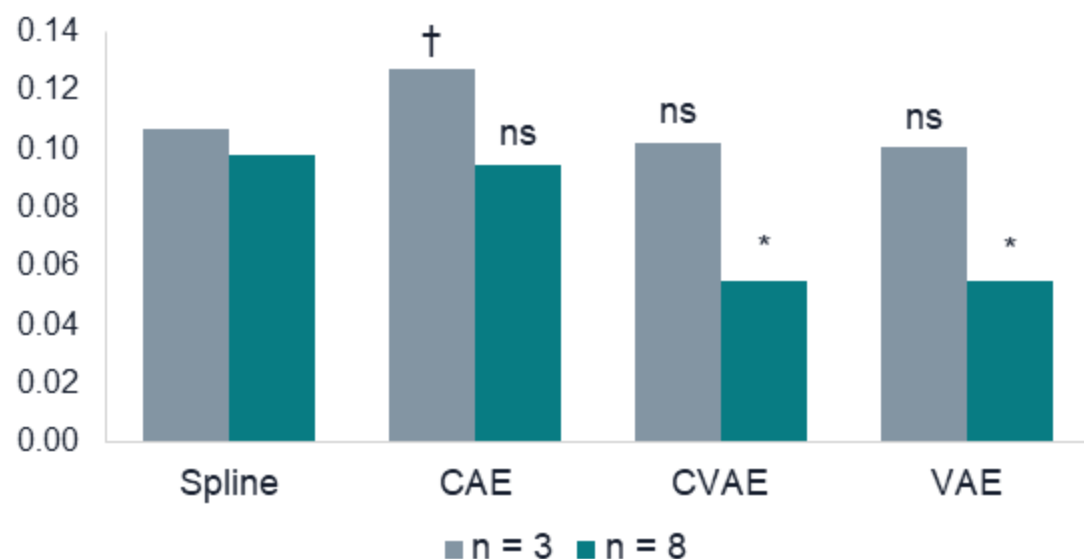
Illustration: subject01, n=8. VAE/CVAE beat spline on all three feature metrics at both capacities (BH and Holm-140). VAE and CVAE give comparable results.

Full-trajectory generation supports VAE and CVAE

Compare generated and held-out path sets: between-set distances plus within-set variability.

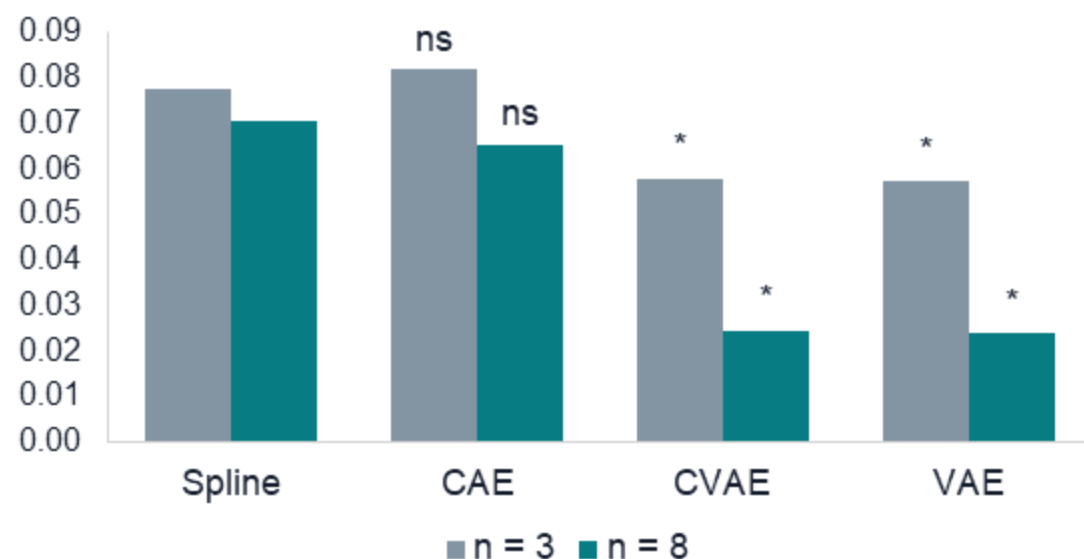
Full-path energy (cm)

vs spline: * BH + Holm; † BH only; ns neither



Full-path MMD squared

vs spline: * BH + Holm; † BH only; ns neither



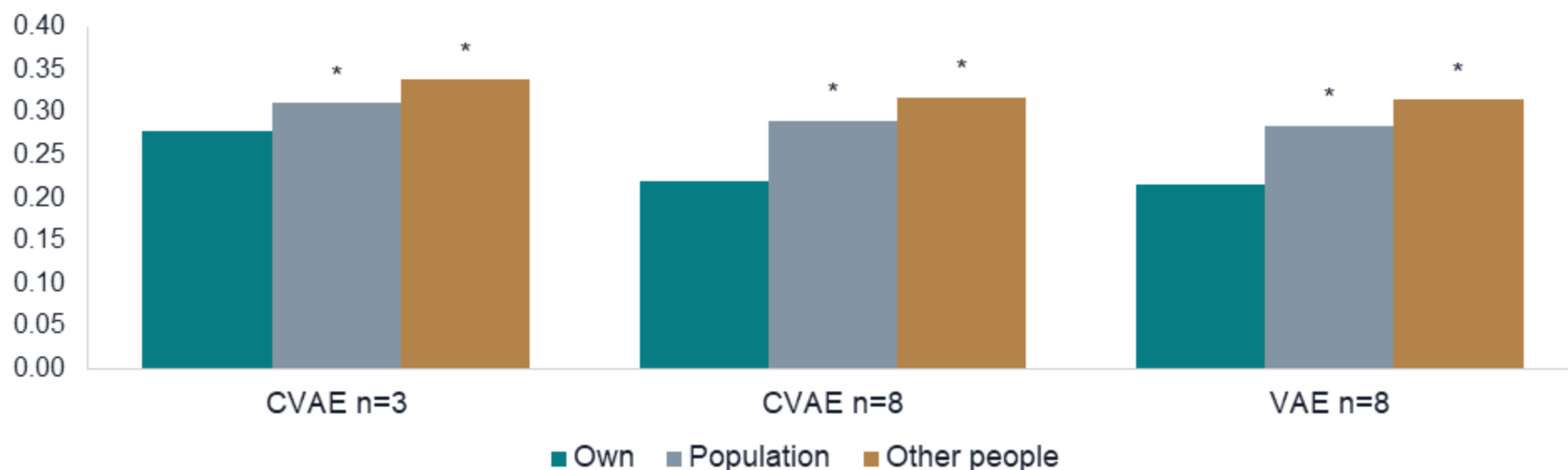
Lower is better. At n=8, both VAE/CVAE beat spline (BH and Holm-80), also with balanced axes. VAE vs CVAE: $q=.672$ and 1.000 . At n=3, results depend on metric and scaling.

The participant's own center improves generation

We change the center while holding decoder, noise and task draws fixed.

Mean feature KS after replacing the center

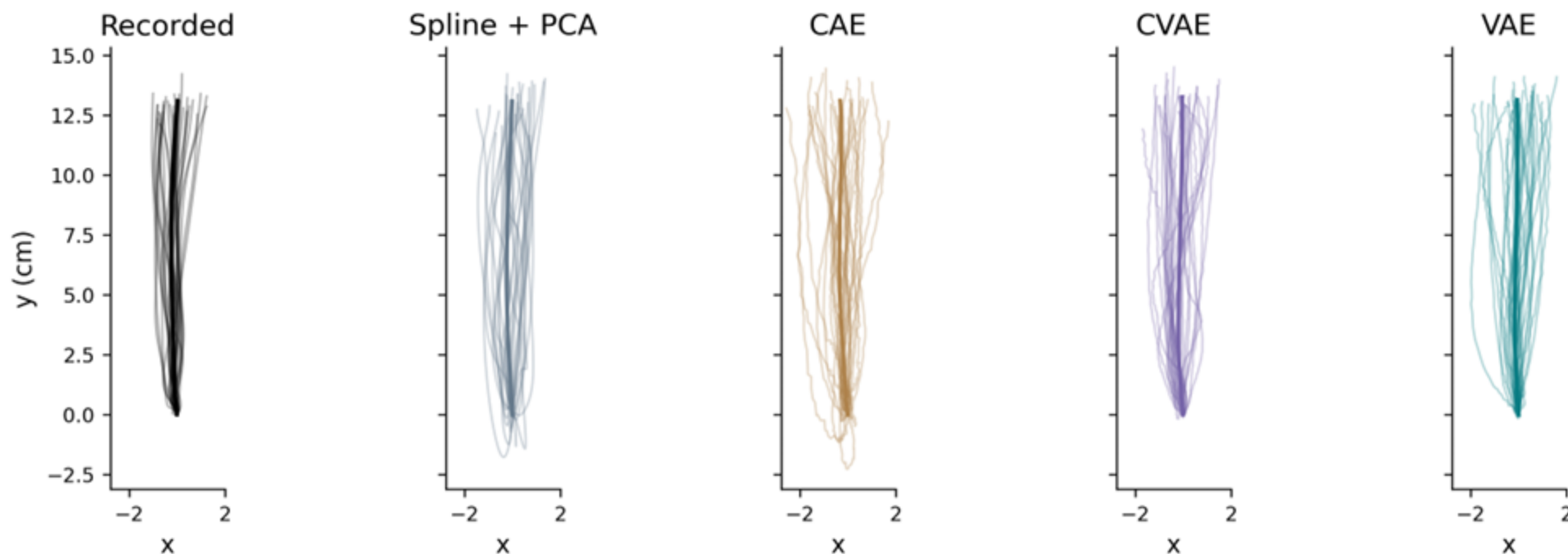
vs own center: * BH-adjusted $p < .05$



Own context improves all tested feature endpoints (BH: VAE $< 1.6e-6$; CVAE $< .000382$). This supports useful personal context within the session.

Dashboard demonstration

VAE $n=8$ is the default. Explore $n=3$, CVAE, CAE and spline + PCA alongside the cohort results.



Explore recorded and generated paths, feature distributions and model benchmarks for the same session.